

SMART INDIA HACKATHON (SIH)

Technical Specification & Solution Report

Problem Statement ID		Category	HARDWARE
Problem Title	AI powered search and Rescue		
Team Name	<u>PUSHPAK</u>	Organization / Ministry	[Insert Ministry Name]

1. Executive Summary

Project Overview

Team Pushpak proposes an AI-Powered Autonomous Search and Rescue Drone designed to accelerate victim detection, enhance situational awareness, and protect first responders in disaster zones. Operating offline without requiring cloud dependencies, the drone provides early detection, multi-sensor data fusion, and real-time mapping for critical emergency situations.

Technical Architecture & Core Capabilities

- Hardware & Propulsion: Powered by a lightweight ~1.3 kg frame featuring 920KV brushless motors, 9.5-inch propellers, and a 3S 5400mAh Li-ion battery, delivering a flight endurance of 25–30 minutes, cruising speeds of 30–35 km/h, and top speeds of 50–55 km/h.
- Multi-Sensor Payload: Integrates high-res RGB visual cameras, thermal cameras for night/heat-signature detection, Ground Penetrating Radar (GPR) to find trapped victims, LiDAR, barometer, IMU, and GPS. It also includes a microphone array and onboard speaker for two-way audio communication.
- Autonomous & Edge Intelligence: Runs on Pixhawk Cube Orange with edge computing (using ROS 2, PX4/ArduPilot, SLAM, PyTorch, OpenCV, and YOLO). It provides GPS-denied navigation, real-time obstacle avoidance, hazard classification (fire, smoke, debris, chemical leaks), and automated confidence scoring.



2. Problem Definition & Scope

2.1 Core Problem Analysis

1. Inefficient & Delayed Victim Location

- The Gap: Traditional Search and Rescue (SAR) operations rely heavily on manual search teams or manned helicopters. High-stress, post-disaster environments (debris, floods, structural collapses) obscure victims and make physical access extremely slow and hazardous.
- Impact: Delays in locating trapped or injured individuals significantly reduce survival rates during the critical "golden hours" following a disaster.

2. Environmental Hazards & Ground Responder Risks

- The Gap: Responders entering disaster zones face immediate threats such as gas/chemical leaks, structural instability, fire, exposed electrical lines, and landslides.
- Impact: First responders often risk their own lives without prior situational awareness of the hazards lurking inside the search zone.

2.2 Project Scope & Boundaries

1. In-Scope Features & Deliverables

- Autonomous Navigation & SLAM: Onboard path planning, obstacle avoidance, and capability to navigate in GPS-denied environments using Visual-Inertial Odometry and SLAM (Simultaneous Localization and Mapping).
- Multi-Sensor Data Fusion: Simultaneous processing of RGB visual feeds, thermal heat signatures, Ground Penetrating Radar (GPR) surface scans, LiDAR, and IMU data.
- Edge AI & Hazard Detection: Real-time on-device inference (PyTorch, OpenCV, YOLO) for victim detection and identification of active hazards (fire, smoke, debris, damaged structures, exposed power lines, chemical leaks).
- Communication & Audio Interaction: Onboard speaker and microphone array for two-way voice alerts/communication with survivors in disaster areas.

3. Proposed Solution Architecture

3.1 System Architecture

1. Multi-Sensor Perception Layer (Payload & Data Acquisition)

Captures environmental and victim data across multiple spectrums:

- Visual & Thermal Sensing: High-res RGB camera for daytime visual mapping; thermal camera for night operations and detecting body heat signatures.
- Sub-Surface Sensing: Ground Penetrating Radar (GPR) camera to scan beneath structural debris and identify trapped victims.
- Spatial & Positioning: LiDAR for precise 3D surface mapping and obstacle detection, supported by barometer, IMU tracking (roll, pitch, yaw), and GPS positioning.

3.2 Tech Stack Justification

• Frontend / UI: Core Framework & Architecture

- Recommended Technology: React / Next.js or Vue.js (Web-based Dashboard Stack)
- Justification:
 - Low-Latency Rendering: Disaster response demands high-performance UI components that can render live multi-stream video feeds (RGB and Thermal) without lag or UI freezes

• Backend / Logic Processing: 1. Programming Languages & Core Runtime

- Technologies: Python, C++
- Justification:
 - C++: Critical for high-performance, low-latency execution of flight control, low-level driver integration, motor control, and real-time sensor loops.
 - Python: Used for rapid deployment of computer vision models, AI inference, data manipulation, and high-level workflow orchestrations.

• Database / Storage: . Onboard Edge Storage (On-Drone Data Capture)

- Technology: Local Embedded Storage (High-Speed NVMe / MicroSD via Companion Computer)
- Role & Functionality:
 - Raw Sensor Buffering: High-speed local write storage for raw high-resolution RGB visual streams, thermal feeds, and Ground Penetrating Radar (GPR) subsurface scans.
 - Telemetry & Flight Logs: Continuous local logging of IMU readings, GPS coordinates, power/battery health metrics, and SLAM spatial mapping trajectories.
 - Model Weights & Local Cache: Local storage of pre-trained Edge AI weights (YOLO, object recognition models) for low-latency offline inference.

• Hardware Components (If applicable): Core Propulsion & Airframe

- Brushless Motors: \$6 \times 920 brushless motors providing reliable thrust and speed control.
- ESC (Electronic Speed Controllers): Matched ESCs paired with motors for precise RPM management and responsive flight maneuvers.
- Propellers: 9.5-inch propellers with a 5-inch pitch, selected to ensure responsive handling, flight stability, and thrust control.
- Airframe Assembly: Lightweight, rugged airframe maintaining an overall empty drone weight of approximately 1.3 kg (including battery pack)



4. Functional Specifications & Modules

• **Module 1: Data Ingestion & Validation** – For Project Pushpak (Smart India Hackathon 2026), Module 1: Data Ingestion & Validation acts as the primary data pipeline layer. It handles high-throughput ingestion from heterogeneous onboard hardware sensors, performs low-latency validation and cleaning, and prepares unified streams for downstream Edge AI and SLAM processing.

Module 1: Data Ingestion & Validation Architecture

1. Multi-Stream Ingestion Layer (Input Sources)

Ingests multi-modal data streams running at varying sample rates directly onboard the companion computer:

- High-Bandwidth Video Feeds: Continuous dual-camera streaming (High-Res RGB + Thermal visual feeds) via OpenCV/GStreamer nodes

• **Module 2: Analytics / Core Logic Engine** – Multi-Modal AI Detection Engine

Executes parallel computer vision and signal processing pipelines using lightweight deep learning models (YOLO, PyTorch, OpenCV):

- Visual & Thermal Victim Detection: Scans high-resolution RGB feeds for human forms/trapped bodies while correlating them with thermal heat signatures to detect living survivors in low-visibility or night conditions

• **Module 3: Visualisation & Alerts** –Real-Time Command Dashboard (UI Visualizations)

Provides ground operators with live situational awareness through a multi-view dashboard interface:

- Dual Live Video Stream Display: Renders simultaneous feeds from high-resolution RGB visual cameras and thermal imaging streams to track operations across day and night conditions.
- Interactive GIS Disaster Map: Overlays real-time drone positioning, flight telemetry, and path vectors onto dynamic map layouts

5. Feasibility, Security & Scalability

5.1 Technical & Economic Feasibility

Hardware Readiness & Integration

- Proven off-the-shelf Components: Built using standardized components (Pixhawk Cube Orange, \$920\text{ KV}\$ motors, 3S \$5400\text{ mAh}\$ battery, lightweight carbon frame) that are readily available in the market, keeping prototype complexity low.
- Multi-Modal Payload Compatibility: Combining lightweight payloads—RGB visual, thermal sensors, LiDAR, and Ground Penetrating Radar (GPR)—holds a total airframe mass of \$\approx 1.3\text{ kg}\$, retaining high agility without overwhelming motor thrust limits.

5.2 Security & Data Privacy

- On-Device Edge Processing (Privacy by Design): Raw camera and thermal streams are processed locally on the companion computer. Live video is converted into anonymized bounding boxes, features, and numerical spatial coordinates rather than streaming unencrypted raw footage continuously over cloud networks.
- Automated PII Redaction & Face Anonymization: AI vision algorithms automatically blur or redact Personally Identifiable Information (PII)—such as face geometry or license plates—of bystanders and non-victims before transmitting feeds to ground control

5.3 Scalability Model

- Swarm Orchestration: The backend control framework (built on ROS 2 and PX4) supports multi-agent operations. Multiple Pushpak drones can be deployed simultaneously to divide large disaster areas into distinct flight grids for rapid parallel mapping.
- Distributed Command Node: The Command Center Dashboard can ingest live telemetry, dual video streams, and hazard updates from multiple active drones onto a single synchronized GIS interface

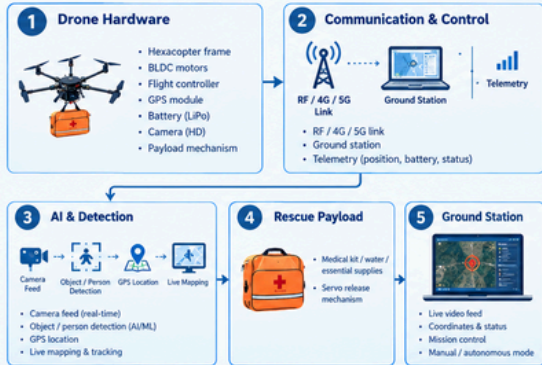


6. Implementation Roadmap & Timeline

Implementation & Roadmap

Search & Rescue Hexacopter Drone System

System Architecture / Implementation Flow



Roadmap



Goal: Fast detection → Accurate location → Safe payload delivery

7. References & Data Sources

Primary References Cited in Slide Deck

- UAV Coach — Search & Rescue Drones

 - Domain: Public safety and sUAS operational standards.
 - Relevance: Technical guidance on aerial reconnaissance workflows, payload selection, and deployment logistics for emergency responders.
- LightPath — Drone Thermal Cameras

 - Domain: Thermal optics and infrared sensor integration.
 - Relevance: Specifications for night-vision, thermal heat signature tracking, and payload weight trade-offs.
- Industrial Batteries — Energy-Efficient Batteries

 - Domain: Flight power systems and battery management.
 - Relevance: Informs the trade-offs regarding battery weight vs. flight time (balancing 3S \$5400\text{ mAh}\$ capacity with a 25–30 min endurance goal).
- Teledyne FLIR — Can Thermal See Through Walls?

 - Domain: Infrared sensor physics and limitations.
 - Relevance: Clarifies that thermal cameras detect surface heat signatures rather than penetrating solid walls, establishing the technical necessity for combining thermal with Ground Penetrating Radar (GPR) for debris penetration.
- JOUAV — Drone Types Guide

 - Domain: Unmanned Aerial Vehicle airframe architectures.
 - Relevance: Comparative analysis of multi-rotor quadcopters vs. fixed-wing drones for stability, payload capacity, and maneuverability in collapsed or constrained disaster zones.